# Written 2

- Due Mar 8 at 10pm
- Points 105
- Questions 18
- Available until Mar 9 at 10pm
- Time Limit None

# Instructions

Read chapter 2 of the textbook and complete the following exercises:

- 2.9
- 2.18
- 2.19
- 2.24
- 2.25
- 2.28
- 2.29
- 2.36

You may type your work into the boxes in the quiz or upload individual images or PDFs for each answer.  You may not upload a single PDF or PDFs that cover multiple answers.  The TAs have complained about the collective disorganization of submissions in past semesters and begged the instructores for this respite (from single PDF uploads).

You must show all work.  Answers without work will receive no credit.  Answers with insufficient work will receive little to no credit

Please keep your work neat and organized.  The TAs will penalize your grade if they are forced to expend undue effort to decipher cryptic submissions.

This quiz was locked Mar 9 at 10pm.

## Attempt History

| | Attempt | Time | Score |
|---|---|---|---|
| **LATEST** | Attempt 1 | 7 minutes | 100 out of 105 |

ⓘ Correct answers are hidden.

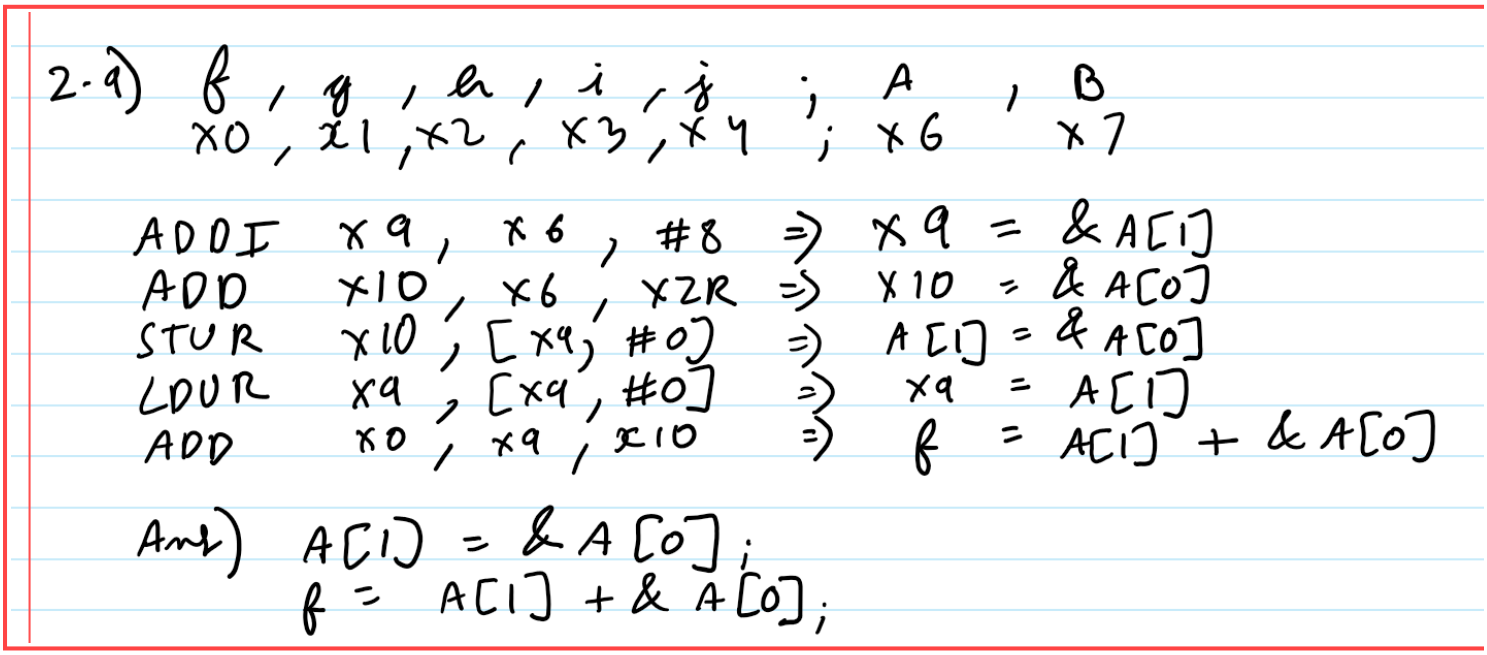
Score for this quiz: 100 out of 105
Submitted Mar 8 at 9:57pm
This attempt took 7 minutes.

Question 1

10 / 10 pts

2.9

Your Answer:




Question 2

5 / 5 pts

2.18.1

Your Answer:

2.18) X10 = 0x0000 0000 AAAA AAAA
X11 = 0x1234 5678 1234 5678

1) LSL X12, X10, #4 ; Shifting X10 by 4 bits

=> X12 = 0x0000 000A AAAA AAA0

ORR X12, X12, X11

0x0000 000A AAAA AAA0 [A = 1010]
OR 0x 1234 5678 1234 5678
0x 1234 567A BABE FEF8

Ans) X12 = 0x1234 567A BABE FEF8

Question 3
5 / 5 pts

2.18.2

Your Answer:

2) LSL X12, X11, #4 ; Shift left 4 bits
X11 = 0x1234 5678 1234 5678

Ans) 0x2345 6781 2345 6780

Question 4
4 / 5 pts

2.18.3

Your Answer:

3) LSR X12, X10, #3

X10 = 0x0000 0000 AAAA AAAA

1010 1010 1010 1010 1010 1010 1010

Right 3 ⇒ 0001 0101 0101 0101 0101 0101 0101

HEX ⇒ 1 5 5 5 5 5 5

Ans) ∴ X12 = 0x0000 0000 1SSS SSSS

0x545

Question 5

10 / 10 pts

2.19

Your Answer:

2-19) Extract bits 16 ... 11 from X10

Put them in bits 31 ... 26 of X11

Ans) 1) LSL X12, X10, #11

2) ANDI X12, X12, #0x3F

3) LSL X12, X12, #26

4) ANDI X11, X11, #0xF03FFFFF

5) ORR X11, X11, X12

Question 6

4.5 / 5 pts

2.24.1

Your Answer:

should be Conditional Branch type.

Question 7
4.5 / 5 pts

2.24.2

Your Answer:

Possible in 3 instructions: SUBIS Rn, Rn, #1 B.PL loop ADDI Rn, Rn, #1

Question 8
5 / 5 pts

2.25.1

Your Answer:

Question 9
5 / 5 pts

2.25.2

Your Answer:

Question 10
5 / 5 pts

2.25.3

Your Answer:

Question 11
5 / 5 pts

2.25.4

Your Answer:

Question 12
5 / 5 pts

2.25.5

Your Answer:

Question 13
5 / 5 pts

2.25.6

Your Answer:

Question 14
2.5 / 5 pts

2.25.7

Your Answer:

X2 doesn't make sense here. If it's X1, the double decrement causing the loop to run N-2 times instead of N.

Question 15
4.5 / 5 pts

2.28

Your Answer:

int i = 0;

Question 16
10 / 10 pts

2.29

Your Answer:

Question 17
5 / 5 pts

2.36.1

Your Answer:

Question 18
5 / 5 pts

2.36.2

Your Answer:

Quiz Score: 100 out of 105